Initial Post

Display replies in nested form

Settings

Initial Post

by Lauren Pechey - Sunday, 24 May 2026, 12:48 PM

Initial Post: Risks and Security Challenges in Industry 4.0

The term *Industry 4.0* refers to the fourth industrial revolution, where advanced digital technologies are integrated into industrial and business operations to improve automation, efficiency, communication, and decision-making. According to Kovaitė and Stankevičienė (2019), Industry 4.0 involves the use of technologies such as Artificial Intelligence (AI), cloud computing, robotics, the Internet of Things (IoT), and big data analytics to create highly connected and intelligent systems. Two examples of Industry 4.0 technologies are IoT-enabled smart factories and autonomous robots. Smart factories use sensors and connected devices to monitor production systems in real time, while autonomous robots can perform repetitive industrial tasks with minimal human intervention (Kovaitė and Stankevičienė, 2019).

The authors identify several risks associated with Industry 4.0, including cybersecurity threats, privacy concerns, technological dependency, and operational disruptions. One real-world example is the 2021 ransomware attack on the Colonial Pipeline, where attackers disrupted fuel distribution systems by exploiting weaknesses in digital infrastructure. This demonstrates the cybersecurity risks created by interconnected systems. Another example is the 2017 Equifax Data Breach, where sensitive information belonging to millions of users was exposed due to poor security controls, highlighting privacy and data protection risks in digital business environments.

Several journal articles support the arguments made by Kovaitė and Stankevičienė regarding the opportunities and risks of Industry 4.0. Xu, Xu and Li (2018) argue that Industry 4.0 improves productivity and operational efficiency but increases organisational exposure to cyberattacks due to interconnected networks and cloud-based systems. Similarly, Lasi et al. (2014) explain that digital transformation changes traditional business models and creates new security and management challenges. In addition, Frank, Dalenogare and Ayala (2019) found that while Industry 4.0 technologies provide strategic and economic benefits, organisations often struggle with implementation risks, lack of cybersecurity readiness, and technological complexity. These studies collectively support the view that businesses must adopt effective risk management and security frameworks to safely benefit from digital transformation.

References:

Frank, A.G., Dalenogare, L.S. and Ayala, N.F. (2019) 'Industry 4.0 technologies: Implementation patterns in manufacturing companies', *International Journal of Production Economics*, 210, pp. 15–26. Available at: **https://doi.org/10.1016/j.ijpe.2019.01.004**

Kovaitė, K. and Stankevičienė, J. (2019) 'Risks of digitalisation of business models during industry 4.0', *Journal of Business Economics and Management*, 20(6), pp. 1079–1098. Available at: **https://doi.org/10.3846/jbem.2019.11217**

Lasi, H., Fettke, P., Kemper, H.G., Feld, T. and Hoffmann, M. (2014) 'Industry 4.0', *Business & Information Systems Engineering*, 6(4), pp. 239–242. Available at: **https://doi.org/10.1007/s12599-014-0334-4**

Xu, L.D., Xu, E.L. and Li, L. (2018) 'Industry 4.0: state of the art and future trends', *International Journal of Production Research*, 56(8), pp. 2941–2962. Available at: **https://doi.org/10.1080/00207543.2018.1444806**

Maximum rating: -

Permalink **Reply**